CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – IV

ELASTICITY, CREEP & SHRINKAGE

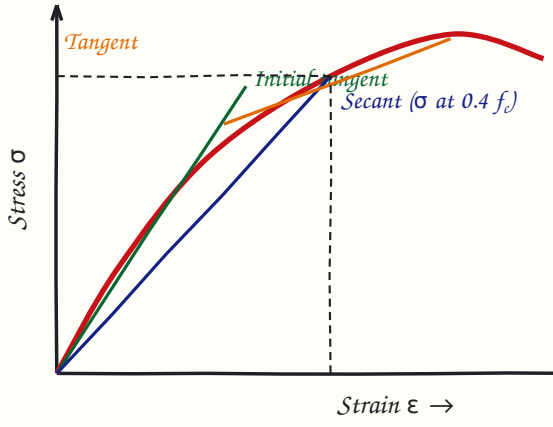
Syllabus covered in this unit

Modulus of elasticity – dynamic modulus of elasticity – Poisson's ratio – creep of concrete – factors influencing creep – relation between creep & time – nature of creep – effects of creep – shrinkage – types of shrinkage

1. ELASTICITY OF CONCRETE

- Concrete is **not truly elastic** : stress–strain curve is **curved (non-linear) from the beginning** because of its heterogenous (aggregate–paste–ITZ) nature, micro-cracking and creep. It is approximately elastic at low stress (< 30–40 % f_c).
- Elastic properties are needed for **deflection, crack width, stress in steel (modular ratio), prestress loss, column shortening, structural analysis.**
- Strains in concrete = elastic + creep + shrinkage + thermal.

1.1 Modulus of elasticity



Types of modulus of elasticity
- **Initial tangent** – slope at origin
- **Tangent** – slope at a given σ
- **Secant** – slope of line from origin to a point (≈ 0.33–0.4 f_c) – used in design
- **Chord** – slope between two points (σ = 50 μ strain to 0.4 f_c)

IS 456 : $E_c = 5000 \sqrt{f_{ck}}$ (N/mm^2)

Modulus of elasticity (Young's) E = stress / strain

Secant modulus (short-term static) is used in design (IS 456 cl. 6.2.3.1) :

$E_c = 5000 \sqrt{f_{ck}}$ N/mm^2 (ACI : $4700 \sqrt{f'_c}$; BS : $20 + 0.2\, f_{cu}$ kN/mm^2)

e.g. M20 → 22 360 ; M25 → 25 000 ; M30 → 27 386 ; M40 → 31 623 N/mm^2

- **Modular ratio** $m = E_s / E_c = 280 / (3\, \sigma_{cbc})$ (IS 456 WSM) ; for long-term including creep,

$E_{ce} = E_c / (1 + \theta)$.

- **Static modulus** = from loading test on 150 × 300 cylinder using compressometer; load–unload 2–3 cycles to remove initial non-linearity; **rate 0.25 N/mm²/s** (IS 516).
- Dependence : increases with f_c (but not proportionally) ; ↑ with age ; ↑ with aggregate stiffness (E_{agg} 20–100 GPa) ; ↓ with ↑ w/c, porosity, moisture (wet concrete slightly higher) ; ↑ with rate of loading.

Concrete grade	f_{ck} (N/mm²)	$E_c = 5000\sqrt{f_{ck}}$ (N/mm²)
M20	20	22 360
M25	25	25 000
M30	30	27 386
M35	35	29 580
M40	40	31 623
M50	50	35 355

- **Composite model** : E_c lies between Reuss (series) and Voigt (parallel) bounds of aggregate (E_a, volume V_a) & paste (E_p, V_p) :

Hirsch's / bounds

Parallel (Voigt) : $E_c = V_p E_p + V_a E_a$ Series (Reuss) : $1 / E_c = V_p/E_p + V_a/E_a$

Hirsch model : $1/E_c = \chi [1/(V_p E_p + V_a E_a)] + (1-\chi)(V_p/E_p + V_a/E_a)$

2. DYNAMIC MODULUS OF ELASTICITY

- Determined from **very small strains** (no creep, no micro-cracking) by **vibration or wave velocity** – corresponds to **initial tangent modulus** ($E_d \approx E_i$) ; Non-destructive (specimens can be re-used).
- **Methods** : (1) **Resonant frequency (longitudinal / flexural / torsional) – IS 516 Part 5 / ASTM C215** ; (2) **UPV** (IS 13311 Part 1).

Dynamic modulus

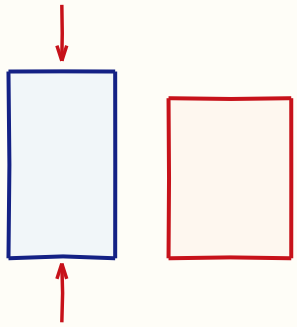
UPV : $E_d = \rho V^2 (1 + \mu)(1 - 2\mu) / (1 - \mu)$ (V = pulse velocity, ρ = density, μ = dynamic Poisson's ratio)

Resonance (longitudinal) : $E_d = 4 n^2 L^2 \rho$ (n = fundamental frequency, L = length) ; Shear $G_d = 4 L^2 n_t^2 \rho$ (torsional)

- **Relation between E_d and E_c (static)** : E_d **is higher** (≈ 20 – 40 % for lean, ~10–20 % rich/high-strength) because no creep and cracking at small strain : $E_c = 1.25 E_d - 19$ (GPa)

(Popovics) ; $E_c = 0.83\ E_d$ (typical). Higher the strength, nearer E_d to E_s.

- Useful for : **quality monitoring**, freeze–thaw damage assessment (**durability factor**), **fatigue**, earthquake analysis, effect of curing.
- Poisson's ratio can be found by combining E_d, G_d : $\mu = (E_d / 2G_d) - 1$.



← loaded shape (shorter, wider)

Poisson's ratio μ = lateral strain / axial strain
- Concrete 0.15 – 0.20 (IS 456 uses 0.2 ; creep…)
- Steel 0.28 – 0.30 ; rubber 0.5
- μ ↑ with stress near failure (> 0.75 f_c) due to micro-cracking (up to 0.5)
- Lower for high-strength & saturated concrete found 0.20 – 0.25 for UPV method

3. POISSON'S RATIO

Poisson's ratio

μ = lateral strain / longitudinal strain ; Relations $E = 2G\ (1 + \mu)$; $K = E / 3(1 - 2\mu)$

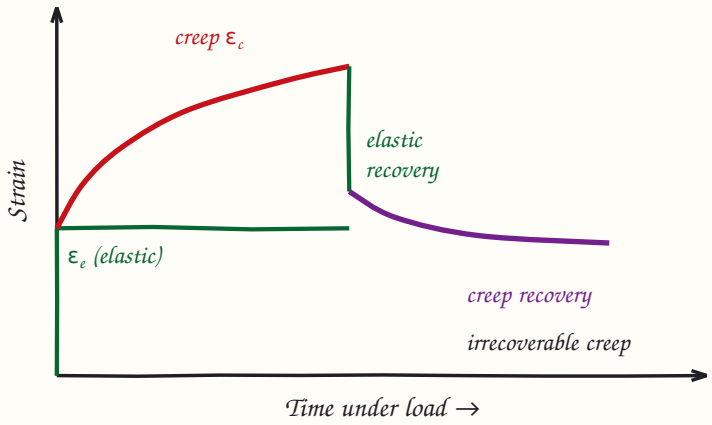
- Concrete : static 0.15 – 0.22 (average 0.2) ; dynamic 0.2 – 0.25 ; **independent of stress up to ~ 0.7 f_c**, then **increases** to ≥ 0.5 → volumetric dilation (due to internal micro-cracking) – signals imminent failure.
- Lower for high-strength concrete, dry concrete ; lightweight concrete ≈ 0.15.
- Used in analysis of slabs, tunnel linings, prestress losses, biaxial stress & pressure vessels.

4. CREEP OF CONCRETE

Definition ★

Creep = **time-dependent increase in strain under sustained (constant) stress**, after the initial elastic strain. It occurs at stress levels even well below the strength.

- Creep strain $\varepsilon_c(t)$ = total strain at t − elastic strain − shrinkage strain (a control specimen).



- Load applied at t_0 → instantaneous elastic strain ε_e
- Strain keeps increasing with time (creep)
- Rate decreases with time
- On unloading : instant elastic recovery + slow partial creep recovery + permanent (residual) deformation

Creep coefficient $\theta = \varepsilon_c / \varepsilon_e$

- **Creep coefficient** θ (φ) = $\varepsilon_c / \varepsilon_e$. **Specific creep (unit creep)** C_u = creep strain / applied stress (10^{-6} per MPa; 40 – 130 × 10^{-6}/MPa typical).
- **Creep recovery** : on unloading, instantaneous elastic recovery (~ same as elastic strain, slightly more as E rises with age) then slow creep recovery (~ 15–25 % of creep, reversible) ; the rest (**irrecoverable**) ~ 75–85 %.
- **IS 456 cl. 6.2.5.1 : creep coefficient** θ (long term) :

Age at loading	Creep coefficient θ
7 days	2.2
28 days	1.6
1 year	1.1

- Creep strain $\varepsilon_{cc} = \theta\, \varepsilon_e = \theta\, \sigma / E_c$; effective modulus $E_{ce} = E_c / (1 + \theta)$.

5. FACTORS INFLUENCING CREEP

1. **Aggregate** : stiffer & denser aggregate → lower creep ; light weight (more creep ~ 1.5×) ; crushed granite < gravel < sandstone ; **volume of aggregate ↑ → creep ↓** (aggregates restrain paste).
2. **Water–cement ratio / paste content** : creep ∝ amount of paste ; lower w/c, better compaction → less creep ; high strength concrete creeps less.
3. **Cement type & fineness** : more C_3S, rapid hardening cement → less creep (stronger at loading) ; low-heat cement more ; blended cement : fly ash/slag reduce long-term creep.
4. **Stress/strength ratio** : linear (**proportional to stress**) up to 0.4–0.5 f_c ; non-linear above (micro-cracking), ≥ 0.75 f_c → creep rupture.
5. **Age at loading** (maturity) : older concrete – less creep (θ at 7 d 2.2 vs 1 yr 1.1).
6. **Relative humidity / curing** : creep decreases as RH rises ; fully-saturated concrete (sealed)

creeps ~ 2–3 × less than dried ; **drying creep (Pickett effect)**.

7. **Temperature** : creep ↑ with temperature (60–70 °C doubles / triples).
8. **Size & shape of member** (volume / surface ratio) : thicker → less creep (**effective thickness**).
9. **Reinforcement** reduces creep by sharing load (compression steel).
10. **Admixtures** : water reducers (lower w/c ↓ creep); accelerators / air entraining ↑ ; silica fume ↓.

6. RELATION BETWEEN CREEP AND TIME

- Creep develops rapidly at first and then slows, tends to a **limiting (ultimate) value** after many years.

Time under load	% of 20-year (ultimate) creep	Remarks
2 weeks	≈ 25 %	Rapid
1 month	≈ 35 – 40 %	
3 months	≈ 50 – 55 %	
1 year	≈ 70 – 75 %	Mostly completed
5 years	≈ 85 – 90 %	
20 – 30 years	100 %	Ultimate

Empirical expressions

Ford & Hansen : $C_t = C_u \cdot t / (a + t)$ (hyperbolic ; $a \approx 10$ days, C_u = ultimate creep)

Ross : $t / C_t = a + b\,t$ ACI 209 : $C_t = [\, t^{0.6} / (10 + t^{0.6}) \,]\, C_u$ (t in days)

Log law : $C_t = a + b \log_{10}(t) - \varepsilon_c \propto \log(t)$

- Example : $C_u = 600 \times 10^{-6}$, ACI 209 at $t = 100$ days : $t^{0.6} = 15.85$; $C_{100} = 15.85/(25.85) \times 600 = 368 \times 10^{-6}$.

7. NATURE (MECHANISM) OF CREEP

- Not completely understood ; theories :
  - **Seepage (moisture-movement) theory** – squeezing out of adsorbed / gel water under stress, to capillary pores.
  - **Viscous flow of cement gel** – sliding of gel sheets / particles (amorphous mass).
  - **Delayed elasticity** – elastic aggregates restraining the creeping paste ; stress transferred slowly (recoverable portion).

- **Plastic flow** (**micro-cracking**, slip at ITZ) – at high stress; **irrecoverable** component.
- **Breakage of bonds & formation of new ones (solidification theory)**.

▶ Basic creep (sealed) + drying creep. Mostly occurs in the **cement paste** ; aggregates do not creep.

Point ★

Creep and shrinkage are similar in cause (**gel water movement**) and are **partly additive** ; they are often studied together — drying creep (Pickett).

8. EFFECTS OF CREEP

▶ **Deflection of beams & slabs** increases (long-term deflection ≈ 2–3 × short-term ; IS 456 Annex C uses factor).

▶ **Loss of prestress** in prestressed concrete (creep + shrinkage + relaxation + elastic shortening ≈ 15–25 %) → account in design.

▶ **Redistribution of stresses** : in columns, creep of concrete transfers load to the steel (steel may yield); in composite members & statically indeterminate structures moments redistribute.

▶ **Column shortening** in tall buildings (differential creep between core and columns).

▶ **Beneficial** : relieves stress concentrations (restraint stress from shrinkage & temperature), reduces cracking in mass concrete & restrained members ; relaxation of internal stresses.

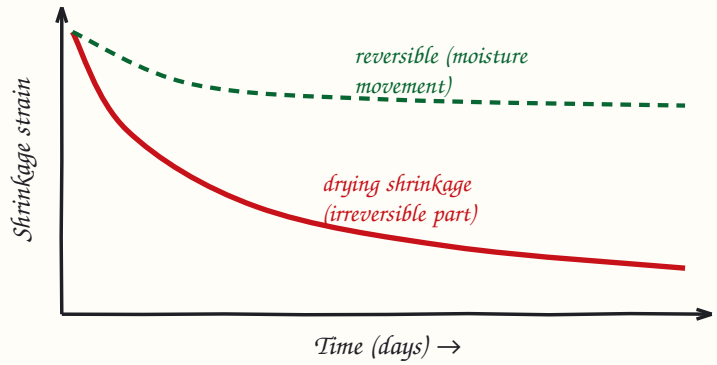
▶ **Effect on crack width, buckling** of slender columns, **arch dam** behaviour, bridges (camber loss).

▶ Remedies : lower w/c, ↑ aggregate stiffness / volume, longer curing & later loading, compression steel, higher strength, steam curing.

9. SHRINKAGE

Definition ★

Shrinkage = **volume reduction** of concrete **independent of external load**, due to loss of water / chemical & thermal changes. Cracks form if restrained.



Moisture movement
- *Drying → shrinkage ; wetting → swelling*
- *Part of drying shrinkage irreversible*
- *Typical 300–800 × 10^{-6}*

(IS 456 : 0.0003 when not available)
- *Rate ∝ exposed surface /volume*
- *Mostly in first few months*

▶ *Typical drying shrinkage* **0.0003 – 0.0008** *(3–8 × 10^{-4}) i.e. 300–800 microstrain (IS 456 cl. 6.2.4 : 0.0003 if unknown). Linear shrinkage per metre = 0.3 – 0.8 mm.*

9.1 Types of shrinkage

Type	Cause	Remarks
Plastic shrinkage	Rapid evaporation of surface water from fresh concrete (rate > 1 kg/m^2/h) before set, exceeding bleeding rate ; hot, windy, dry	Map / diagonal cracks in slabs within 1–6 h ; **prevent by fog spray, wind-breaks, curing compounds, polypropylene fibres**
Drying shrinkage	Loss of **adsorbed gel water** from hardened paste in dry air (RH < 100 %)	Main type ; partly irreversible ; proportional to water content & paste volume
Autogenous shrinkage	**Self-desiccation** – chemical shrinkage of cement hydration (hydrated products smaller than reactants by ~ 10 %), no moisture exchange	Significant for w/c < 0.42, **HPC / silica fume**, at early age
Carbonation shrinkage	$CO_2 + Ca(OH)_2 \rightarrow CaCO_3$; volume reduction (water released) at RH ≈ 50 %	Surface layer, with drying shrinkage
Thermal shrinkage	Cooling after peak hydration temperature (α ≈ 10 × 10^{-6} /°C)	Mass concrete, dams, thick raft ; control by low-heat cement, pre-cooling, pipe cooling

9.2 Factors affecting shrinkage

▶ **Water content** (↑ shrinkage ~ linearly with water content) ; **w/c ratio**, cement content ; **aggregate** (stiff, large, high volume ↓ shrinkage; sandstone, slate ↑; granite, limestone ↓) ; **fineness** & type of cement (RHC, high C_3A more) ; **admixtures** ($CaCl_2$ ↑ 10–50 %, retarders ↑, SP with water cut ↓) ; **relative humidity** (shrinkage 0 at 100 %, 4 × at 40 % RH) ; **size / shape** (V/S ratio – thicker slower) ; **curing** (longer curing delays but does not reduce) ;

reinforcement (restrains ~ 10 – 40 %) ; **carbonation** .

9.3 Control of shrinkage cracking

- Minimum water, proper aggregate grading with max size, **good curing 7–14 days** ; **contraction / control joints** (spacing 3–6 m in slabs ; < 4.5 m in walls), **temperature & shrinkage steel** (0.15 % for HYSD in slabs, IS 456 cl. 26.5.2.1), **shrinkage-compensating cement** (Type K, expansive – ettringite), **shrinkage reducing admixtures (SRA)**, fibres, fly ash, avoiding delay.

10. CREEP vs SHRINKAGE – COMPARISON

Aspect	Creep	Shrinkage
Load dependence	Under **sustained stress**	**Independent** of load
Cause	Gel water seepage, viscous flow, micro-cracking	Loss of water, thermal, chemical changes
Reversibility	Partly recoverable on unloading	Partly reversible on re-wetting
Measured by	Loaded minus unloaded (control) specimens	Unloaded specimen, strain gauge / dial gauge
Typical magnitude	2–3 × elastic strain (θ = 1.1–2.2)	300 – 800 × 10^{-6}
Time	Continues for years	Mostly in first months–1 yr
Effect	Deflection, prestress loss, stress redistribution	Cracks, warping, prestress loss

11. MOISTURE & THERMAL MOVEMENT

- **Moisture movement** : swelling (on wetting, expansion ≈ 100–200 × 10^{-6}) and shrinkage (drying) – cyclic wetting/drying causes fatigue ; **irreversible shrinkage** ≈ 30–60 % of total on first drying.
- **Coefficient of thermal expansion** of concrete : 7 – 13 × 10^{-6} /°C (typical 10 × 10^{-6}) ≈ steel (12 × 10^{-6}) → no thermal stress between steel & concrete ; depends on aggregate (quartz ↑, limestone ↓).
- **Thermal stress** if restrained : $\sigma = E \alpha \Delta T / (1 - \mu)$ (ΔT = 20 °C, E = 25 GPa, $\alpha = 10^{-5}$ → σ ≈ 5 MPa > tensile strength → cracks).

12. SOLVED NUMERICALS

Problem 1 – modulus of elasticity

Find E_c for M35 concrete, and short-term shortening of a 3 m column carrying 5 MPa.

Solution

$E_c = 5000\sqrt{35} = 5000 \times 5.916 =$ **29 580 N/mm²** ; $\varepsilon = \sigma/E = 5/29\,580 = 1.69 \times 10^{-4}$; $\Delta L = 3000 \times 1.69\times10^{-4} =$ **0.51 mm**

Problem 2 – creep deformation

Column 3 m long carries stress 6 MPa, M30 concrete, loaded at 28 days. Find long-term creep strain & total shortening ($\theta = 1.6$).

Solution

$E_c = 5000\sqrt{30} = 27\,386$ MPa ; $\varepsilon_e = 6/27\,386 = 2.19 \times 10^{-4}$; $\varepsilon_{creep} = \theta\,\varepsilon_e = 1.6 \times 2.19\times10^{-4} = 3.51 \times 10^{-4}$

Total $= 5.70 \times 10^{-4} \rightarrow \Delta L = 3000 \times 5.70\times10^{-4} =$ **1.71 mm** (creep portion 1.05 mm) ; $E_{ce} = 27\,386 / 2.6 = 10\,533$ MPa

Problem 3 – dynamic modulus

UPV = 4.4 km/s, density 2400 kg/m³, $\mu = 0.2$:

Solution

$E_d = 2400 \times (4400)^2 \times (1.2)(0.6)/(0.8) = 2400 \times 1.936 \times 10^7 \times 0.9 =$ **4.18 × 10¹⁰ Pa = 41.8 GPa** (static $E_c \approx 0.8$–$0.9\,E_d \approx 35$ GPa)

Problem 4 – shrinkage

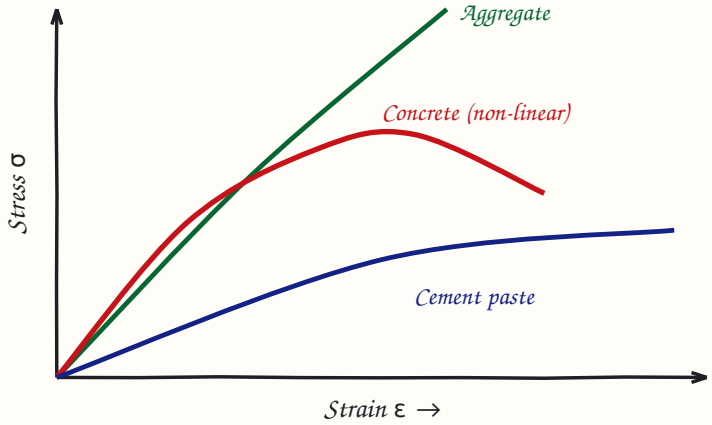
Slab 6 m long, shrinkage strain 0.0003 fully restrained, $E_c = 25\,000$ MPa : tensile stress $\sigma = E\,\varepsilon = 25\,000 \times 0.0003 =$ **7.5 MPa** (> f_{ct}, hence cracks ; creep relieves ~ 50 – 70 %) ; free shortening $= 6000 \times 0.0003 =$ **1.8 mm**.

13. IMPORTANT QUESTIONS (EXAM)

1. Define modulus of elasticity & explain types ; give IS 456 expression.
2. Explain dynamic modulus of elasticity and how it is determined. Relate it with static modulus.

3. Define creep. Explain factors influencing creep & the creep–time relationship.
4. Explain nature (mechanisms) and effects of creep on structures.
5. What is shrinkage ? Explain different types of shrinkage & factors affecting.
6. Compare creep and shrinkage. How cracking due to shrinkage is controlled ?
7. Define Poisson's ratio & state values for concrete.

14. STRESS–STRAIN BEHAVIOUR OF CONCRETE



- Aggregate and paste are each almost linear-elastic ; concrete is **curved** due to ITZ micro-cracking
- IS 456 design curve : parabola up to $\varepsilon = 0.002$ then constant to $\varepsilon_{cu} = 0.0035$
- Peak stress at $\varepsilon \approx 0.002$; HSC is steeper and more brittle
- Descending branch only in stiff machines

- Region 1 ($\sigma < 0.3\ f_c$) : nearly linear, bond cracks (pre-existing) stable ; Region 2 (0.3–0.5) : slight curvature, bond cracks grow ; Region 3 (0.5–0.75) : mortar cracks, strain increases faster ; Region 4 (> 0.75 f_c) : unstable cracking → failure at **$\varepsilon \approx 0.002$–$0.0035$** .
- **Ultimate strain** : 0.0035 (IS 456), 0.003 (ACI); HSC lower (0.0025–0.003), confinement increases ductility (stirrups, spiral).
- **Design stress block** : 0.45 f_{ck} (WSM bending) ; limit state : 0.446 f_{ck} over 0.42 x_u.

15. TESTS FOR CREEP & SHRINKAGE

- **Creep test (ASTM C512 / IS 1343)** : 150 × 300 mm cylinders in a **creep rig** (spring-loaded frame) under constant stress ≤ 40 % f_c ; companion unloaded specimens measure shrinkage; temperature 23 ± 2 °C, RH 50 ± 4 % ; strain measured with Demec (mechanical) gauge / LVDT / vibrating-wire gauges at set time intervals (1 d, 3 d, 7 d, 14 d, 28 d, 3 m, 6 m, 1 y).
- **Shrinkage test (IS 1199 / ASTM C157)** : prism 75 × 75 × 285 mm with gauge studs ; length comparator reading after 28-day moist curing then drying at 50 % RH – strain = $\Delta L / L$. Reading at 4, 7, 14, 28, 56 days etc.
- **Moisture movement** : prisms alternating wet–dry.

16. RHEOLOGICAL MODELS (BRIEF)

- **Maxwell model** (spring + dashpot in series) – stress relaxation. **Kelvin–Voigt** (parallel) – delayed elasticity / creep recovery. **Burgers model** (Maxwell + Kelvin in series) fits creep well : instantaneous elastic + delayed elastic + viscous flow.
- $\varepsilon(t) = \sigma/E_1 + (\sigma/E_2)(1 - e^{-t/\tau}) + \sigma\, t / \eta$ (Burgers).
- **Relaxation** : decrease in stress at constant strain over time – important in prestressing (concrete relaxation ≈ 20–30 % of creep stress loss).

17. DESIGN PROVISIONS (CREEP, SHRINKAGE)

- **Long-term deflection (IS 456 cl. 23.2 & Annex C)** : $\delta_{total} = \delta_{shrinkage} + \delta_{creep+instantaneous}$; deflection ≤ span/250 (total) and span/350 or 20 mm (after finishes) ; use effective modulus E_{ce}.
- **Prestress losses (IS 1343)** : creep loss = $\varepsilon_{cc}\, E_s$; ε_{cc} = ultimate creep strain per MPa (= 0.000 044 / MPa typical) × stress at steel level ; shrinkage loss = $\varepsilon_{cs}\, E_s$ (ε_{cs} = 0.0003 pre-tension, 0.0002 / $\log_{10}(t+2)$ post-tension) ; relaxation of steel 3–9 %.
- **Reinforcement for shrinkage & temperature** : minimum 0.15 % (Fe 415, 500) – 0.12 % of gross area in slabs (IS 456 cl. 26.5.2.1) ; spacing ≤ 5 d or 450 mm.
- **Joints** : expansion joints for structures > 45 m ; control joints spacing 3–4.5 m in floor slabs on grade.

18. MORE NUMERICALS

Problem 5 – prestress loss due to creep

Stress at steel level = 7 N/mm^2, specific creep = 4.4×10^{-5} per N/mm^2, $E_s = 2 \times 10^5$ N/mm^2 → creep loss = $7 \times 4.4\times10^{-5} \times 2\times10^5$ = **61.6 N/mm^2** (≈ 4.4 % of 1400 N/mm^2 initial stress).

Problem 6 – equivalent modulus ratio

Concrete M25, $E_s = 2\times10^5$. Short-term modular ratio m = 200 000 / 25 000 = **8** ; with θ = 1.6 → long-term m = 8 × 2.6 = **20.8** (shows steel stress rise in columns with time).

Problem 7 – thermal strain

A 30 m roof slab, $\alpha = 10 \times 10^{-6}/°C$, temperature rise 25 °C : free expansion = $30\,000 \times 10^{-5} \times 25 =$ **7.5 mm** – provide expansion joint.

19. ONE-PAGE SUMMARY

Property	Symbol / value	Range for normal concrete
Static modulus	$E_c = 5000\sqrt{f_{ck}}$	20 – 40 GPa
Dynamic modulus	$E_d = \rho V^2(\dots)$	25 – 50 GPa (≈ 1.1–1.4 E_c)
Poisson's ratio	μ	0.15 – 0.22
Creep coefficient	$\theta = \varepsilon_c/\varepsilon_e$	1.1 – 2.2 (IS 456)
Drying shrinkage	ε_{sh}	$300 – 800 \times 10^{-6}$
Thermal expansion	α	$7 – 13 \times 10^{-6}$ /°C
Ultimate strain	ε_{cu}	0.0035 (IS 456)

Quick revision – Unit IV ★

- $E_c = 5000\sqrt{f_{ck}}$; $E_d > E_s$; $\mu = 0.15–0.2$; $E_d = \rho V^2(1+\mu)(1-2\mu)/(1-\mu)$.
- Creep θ : 2.2 (7 d), 1.6 (28 d), 1.1 (1 yr) ; creep ∝ stress up to 0.4 f_c.
- Shrinkage : plastic, drying, autogenous, carbonation (+ thermal) ; IS 456 : 0.0003.